

打包全过程

- $X_m[j]$:第m个样本的第j个特征
- 假设有t维特征，n张图像（即一个样本是一个 $t \times 1$ 的向量）
- $W_{t' \times t}$: 权重矩阵，t'表示本层神经元数量，t表示上层神经元（特征数）
- 密文slot: s
- 每张图像取k个特征与其余图像打包，则一个密文中包含s/k个样本（我们可以设batchSize为s/k）

前向传播

每张图像取k个特征，每个密文有s/k个特征，则n张图像我们共需要nk/s个密文

对于一个batch，取以下为一组，一个batch共有t/k个密文

$$\begin{aligned} & X_0[0], X_1[0] \dots X_{s/k-1}[0] | X_0[1], X_1[1] \dots X_{s/k-1}[1] | \dots | X_0[k-1], X_1[k-1] \dots X_{s/k-1}[k-1] \\ & \quad \odot \\ & W_{0,0}, W_{0,0} \dots W_{0,0} | W_{0,1}, W_{0,1} \dots W_{0,1} | \dots | W_{0,k-1}, W_{0,k-1} \dots W_{0,k-1} \\ & \quad = Z_{0,0:k-1} \end{aligned}$$

表示一个batch的前 $k-1$ 个特征与权重矩阵的第0行前 k 列相乘

$$\begin{aligned} & X_0[k], X_1[k] \dots X_{s/k-1}[k] | X_0[k+1], X_1[k+1] \dots X_{s/k-1}[k+1] | \dots | X_0[2k-1], X_1[2k-1] \dots X_{s/k-1}[2k-1] \\ & \quad \odot \\ & W_{0,k}, W_{0,k} \dots W_{0,k} | W_{0,k+1}, W_{0,k+1} \dots W_{0,k+1} | \dots | W_{0,2k-1}, W_{0,2k-1} \dots W_{0,2k-1} \\ & \quad = Z_{0,k:2k-1} \end{aligned}$$

表示一个batch的 $k : 2k-1$ 个特征与权重矩阵的第0行 $k : 2k-1$ 列相乘

:

$$\begin{aligned} & X_0[t-k-1] \dots X_{s/k-1}[t-k-1] | X_0[t-k], X_1[t-k] \dots X_{s/k-1}[t-k] | \dots | X_0[t-1], X_1[t-1] \dots X_{s/k-1}[t-1] \\ & \quad \odot \\ & W_{0,t-k-1}, W_{0,t-k-1} \dots W_{0,t-k-1} | W_{0,t-k}, W_{0,t-k} \dots W_{0,t-k} | \dots | W_{0,t-1}, W_{0,t-1} \dots W_{0,t-1} \\ & \quad = Z_{0,t-k-1:t-1} \end{aligned}$$

表示一个batch的最后 k 个特征与权重矩阵的第0行 $t-k-1 : t-1$ 列相乘

以上为一个batch的图像与权重矩阵的第一行相乘的结果

上述密文进行 $\oplus$ 操作后进行logk次旋转，得到以下结果即第0个隐藏层神经元的输出， Z_m^j 表示第m张图像在第j个神经元的输出

$$Z_0^0, Z_1^0, Z_2^0 \dots Z_{s/k-1}^0 | Z_0^0, Z_1^0, Z_2^0 \dots Z_{s/k-1}^0 | \dots | Z_0^0, Z_1^0, Z_2^0 \dots Z_{s/k-1}^0$$

同理得其余t'-1个神经元的输出如下

$$Z_0^1, Z_1^1, Z_2^1 \dots Z_{s/k-1}^1 | Z_0^1, Z_1^1, Z_2^1 \dots Z_{s/k-1}^1 | \dots | Z_0^1, Z_1^1, Z_2^1 \dots Z_{s/k-1}^1$$

$$Z_0^2, Z_1^2, Z_2^2 \dots Z_{s/k-1}^2 | Z_0^2, Z_1^2, Z_2^2 \dots Z_{s/k-1}^2 | \dots | Z_0^2, Z_1^2, Z_2^2 \dots Z_{s/k-1}^2$$

⋮

$$Z_0^{t'-1}, Z_1^{t'-1}, Z_2^{t'-1} \dots Z_{s/k-1}^{t'-1} | Z_0^{t'-1}, Z_1^{t'-1}, Z_2^{t'-1} \dots Z_{s/k-1}^{t'-1} | \dots | Z_0^{t'-1}, Z_1^{t'-1}, Z_2^{t'-1} \dots Z_{s/k-1}^{t'-1}$$

生成掩码将上述密文进行重组

$$\begin{aligned} m_0 &= [111\dots 111 | 000\dots 000 | 000\dots 000 | \dots | 000\dots 000] \\ m_1 &= [000\dots 000 | 111\dots 111 | 000\dots 000 / \dots | 000\dots 000] \\ &\vdots \\ m_k &= [000\dots 000 | 000\dots 000 | 000\dots 000 | \dots | 111\dots 111] \end{aligned}$$

每个密文乘对应的掩码得以下结果

$$Z_0^0, Z_1^0, Z_2^0 \dots Z_{s/k-1}^0 | Z_0^1, Z_1^1, Z_2^1 \dots Z_{s/k-1}^1 | \dots | Z_0^{k-1}, Z_1^{k-1}, Z_2^{k-1} \dots Z_{s/k-1}^{k-1}$$

$$Z_0^k, Z_1^k, Z_2^k \dots Z_{s/k-1}^k | Z_0^{k+1}, Z_1^{k+1}, Z_2^{k+1} \dots Z_{s/k-1}^{k+1} | \dots | Z_0^{2k-1}, Z_1^{2k-1}, Z_2^{2k-1} \dots Z_{s/k-1}^{2k-1}$$

⋮

$$Z_0^{t-k-1}, Z_1^{t-k-1}, Z_2^{t-k-1} \dots Z_{s/k-1}^{t-k-1} | Z_0^{t-k}, Z_1^{t-k}, Z_2^{t-k} \dots Z_{s/k-1}^{t-k} | \dots | Z_0^{t-1}, Z_1^{t-1}, Z_2^{t-1} \dots Z_{s/k-1}^{t-1}$$

过激活函数之后：

$$A_0^0, A_1^0, A_2^0 \dots A_{s/k-1}^0 | A_0^1, A_1^1, A_2^1 \dots A_{s/k-1}^1 | \dots | A_0^{k-1}, A_1^{k-1}, A_2^{k-1} \dots A_{s/k-1}^{k-1}$$

$$A_0^k, A_1^k, A_2^k \dots A_{s/k-1}^k | A_0^{k+1}, A_1^{k+1}, A_2^{k+1} \dots A_{s/k-1}^{k+1} | \dots | A_0^{2k-1}, A_1^{2k-1}, A_2^{2k-1} \dots A_{s/k-1}^{2k-1}$$

⋮

$$A_0^{t'-1}, A_1^{t'-1}, A_2^{t'-1} \dots A_{s/k-1}^{t'-1} | A_0^{t'-1}, A_1^{t'-1}, A_2^{t'-1} \dots A_{s/k-1}^{t'-1} | \dots | A_0^{t'-1}, A_1^{t'-1}, A_2^{t'-1} \dots A_{s/k-1}^{t'-1}$$

为简便权重矩阵的表示与上一层相同

$$\begin{aligned} &A_0^0, A_1^0, A_2^0 \dots A_{s/k-1}^0 | A_0^1, A_1^1, A_2^1 \dots A_{s/k-1}^1 | \dots | A_0^{k-1}, A_1^{k-1}, A_2^{k-1} \dots A_{s/k-1}^{k-1} \\ &\quad \odot \\ &W_{0,0}, W_{0,0} \dots W_{0,0} | W_{0,1}, W_{0,1} \dots W_{0,1} | \dots | W_{0,k-1}, W_{0,k-1} \dots W_{0,k-1} \\ &\quad = O_{0,0:k-1} \end{aligned}$$

表示隐藏层一 > 输出层的权重矩阵第0行前k列与隐藏层前k个神经元的输出计算的结果

$$\begin{aligned} &A_0^k, A_1^k, A_2^k \dots A_{s/k-1}^k | A_0^{k+1}, A_1^{k+1}, A_2^{k+1} \dots A_{s/k-1}^{k+1} | \dots | A_0^{2k-1}, A_1^{2k-1}, A_2^{2k-1} \dots A_{s/k-1}^{2k-1} \\ &\quad \odot \\ &W_{0,k}, W_{0,k} \dots W_{0,k} | W_{0,k+1}, W_{0,k+1} \dots W_{0,k+1} | \dots | W_{0,k+1}, W_{0,k+1} \dots W_{0,k+1} \\ &\quad = O_{0,k:2k-1} \end{aligned}$$

⋮

$$A_0^{t'-k-1}, A_1^{t'-k-1}, A_2^{t'-k-1} \dots A_{s/k-1}^{t'-k-1} | A_0^{t'-k}, A_1^{t'-k}, A_2^{t'-k} \dots A_{s/k-1}^{t'-k} | \dots | A_0^{t'-1}, A_1^{t'-1}, A_2^{t'-1} \dots A_{s/k-1}^{t'-1} \\ \odot \\ W_{0,t'-k-1}, W_{0,t'-k-1} \dots W_{0,t'-k-1} | W_{0,t'-k}, W_{0,t'-k} \dots W_{0,t'-k} | \dots | W_{0,t'-1}, W_{0,t'-1} \dots W_{0,t'-1} \\ = O_{0,t'-k-1:t'-1}$$

以上共有t'/k个，同上一层， $\oplus$ 后旋转，得到输出层第0个神经元的output， O_m^j 表示第m个样本在第j个神经元的输出

$$O_0^0, O_1^0, O_2^0 \dots O_{s/k-1}^0 | O_0^0, O_1^0, O_2^0 \dots O_{s/k-1}^0 | \dots | O_0^0, O_1^0, O_2^0 \dots O_{s/k-1}^0$$

这样的共有t''个

输出层我们使用softmax函数计算， $p_i = \frac{e^{O^i}}{\sum e^{O^i}}$

对输出层的t''个密文进行指数计算

$$e^{O_0^0}, e^{O_1^0} \dots e^{O_{s/k-1}^0} | \dots | e^{O_0^0}, e^{O_1^0} \dots e^{O_{s/k-1}^0}$$

$$e^{O_0^1}, e^{O_1^1} \dots e^{O_{s/k-1}^1} | \dots | e^{O_0^1}, e^{O_1^1} \dots e^{O_{s/k-1}^1}$$

⋮

$$e^{O_0^{t''-1}}, e^{O_1^{t''-1}} \dots e^{O_{s/k-1}^{t''-1}} | \dots | e^{O_0^{t''-1}}, e^{O_1^{t''-1}} \dots e^{O_{s/k-1}^{t''-1}}$$

对上述t''个密文相加后得

$$\sum_{i=0}^{t''-1} e^{O_0^i}, \sum_{i=0}^{t''-1} e^{O_1^i} \dots \sum_{i=0}^{t''-1} e^{O_{s/k-1}^i} | \sum_{i=0}^{t''-1} e^{O_0^i}, \sum_{i=0}^{t''-1} e^{O_1^i} \dots \sum_{i=0}^{t''-1} e^{O_{s/k-1}^i} | \dots | \sum_{i=0}^{t''-1} e^{O_0^i}, \sum_{i=0}^{t''-1} e^{O_1^i} \dots \sum_{i=0}^{t''-1} e^{O_{s/k-1}^i}$$

计算softmax

$$P_0 = \frac{e^{O_0^0}}{\sum_{i=0}^{t''-1} e^{O_0^i}}, \frac{e^{O_1^0}}{\sum_{i=0}^{t''-1} e^{O_1^i}} \dots \frac{e^{O_1^0}}{\sum_{i=0}^{t''-1} e^{O_{s/k-1}^i}} | \dots | \dots \text{每组都相同}$$

$\frac{e^{O_0^0}}{\sum_{i=0}^{t''-1} e^{O_0^i}}$ 表示样本0在第0个类别的概率，记为 p_0^0

$$\begin{aligned}
P_0 &= [p_0^0, p_1^0, p_2^0 \dots p_{s/k-1}^0 | \dots | p_0^0, p_1^0, p_2^0 \dots p_{s/k-1}^0] \text{每组都相同} \\
P_1 &= [p_0^1, p_1^1, p_2^1 \dots p_{s/k-1}^1 | \dots | p_0^1, p_1^1, p_2^1 \dots p_{s/k-1}^1] \\
&\vdots \\
P_{t''-1} &= [p_0^{t''-1}, p_1^{t''-1}, p_2^{t''-1} \dots p_{s/k-1}^{t''-1} | \dots | p_0^{t''-1}, p_1^{t''-1}, p_2^{t''-1} \dots p_{s/k-1}^{t''-1}]
\end{aligned}$$

对每个样本都需要计算交叉熵损失函数，第m个样本的交叉熵损失 $L_m = - \sum_{j=0}^{t''-1} y_m^j \log(p_m^j)$

首先需要对标签进行编码，明文状态的one-hot编码 $y_m = [y_m^0, y_m^1 \dots y_m^{t''}]$, $y_m^j = 1$ 当且仅当 m 属于类别 j

$$\begin{aligned}
Y_0 &= y_0^0, y_1^0, y_2^0 \dots y_{s/k-1}^0 | y_0^0, y_1^0, y_2^0 \dots y_{s/k-1}^0 | \dots | \\
&\quad \text{每组都相同} \\
Y_1 &= y_0^1, y_1^1, y_2^1 \dots y_{s/k-1}^1 | y_0^1, y_1^1, y_2^1 \dots y_{s/k-1}^1 | \dots | \\
&\vdots \\
Y_{t''-1} &= y_0^{t''-1}, y_1^{t''-1}, y_2^{t''-1} \dots y_{s/k-1}^{t''-1} | y_0^{t''-1}, y_1^{t''-1}, y_2^{t''-1} \dots y_{s/k-1}^{t''-1} | \dots |
\end{aligned}$$

对 P_i 取对数后计算交叉熵损失函数

$$- \sum_{j=0}^{t''-1} (Y^j \odot \log(P^j))$$

得到

$$L = [L_0, L_1 \dots L_{s/k-1} | \dots | L_0, L_1 \dots L_{s/k-1}]$$

组内求和，旋转 $\log_2 s/k$ 可得

$$[\bar{L}, \bar{L}, \bar{L}, \bar{L} \dots \bar{L}, \bar{L}]$$

后向传播

输出层梯度

由于使用交叉熵损失和Softmax激活函数，第m个样本在第j个输出神经元的梯度为： $\delta_m^j = p_m^j - y_m^j$

在密文状态下，第j个输出类别的梯度密文为： $\Delta^j = P^j - Y^j$

展开形式： $\Delta^j = [\delta_0^j, \delta_1^j, \dots, \delta_{s/k-1}^j | \delta_0^j, \delta_1^j, \dots, \delta_{s/k-1}^j | \delta_0^j, \delta_1^j, \dots, \delta_{s/k-1}^j \dots \dots]$

输出层权重梯度

对于权重 $W_{j,l}$ （连接隐藏层第l个神经元到输出层第j个神经元），其梯度为： $\frac{\partial \bar{L}}{\partial W_{j,l}} = \frac{1}{s/k} \sum_{m=0}^{s/k-1} \delta_m^j \cdot A_m^l$

来源：

- $\bar{L} = \frac{1}{s/k} \sum_{m=0}^{s/k-1} L_m$ （平均损失定义）

- 链式法则： $\frac{\partial L_m}{\partial W_{j,l}} = \frac{\partial L_m}{\partial O_m^j} \cdot \frac{\partial O_m^j}{\partial W_{j,l}}$
- 其中 $\frac{\partial L_m}{\partial O_m^j} = \delta_m^j$, $\frac{\partial O_m^j}{\partial W_{j,l}} = A_m^l$

计算前k列权重的梯度 $G_{0,0:k-1}$ 表示输出层的第0个神经元的0~k-1列权重的单样本梯度

$$G_{0,0:k-1} = \Delta^0 \odot A_0 \\ = [\delta_0^0 A_0^0, \delta_1^0 A_1^0 \dots \delta_{s/k-1}^0 A_{s/k-1}^0 | \delta_0^0 A_0^1, \delta_1^0 A_1^0 \dots \delta_{s/k-1}^0 A_{s/k-1}^1 | \dots | \delta_0^0 A_0^{k-1}, \delta_1^0 A_1^{k-1} \dots \delta_{s/k-1}^0 A_{s/k-1}^{k-1}]$$

同理可得

$$G_{0,k:2k-1} = \Delta^0 \odot A_0, G_{0,2k:3k-1} = \Delta^0 \odot A_1 \dots G_{0,t'-k-1:t'-1} = \Delta^0 \odot A_{t'/k-1}$$

将每个G都进行组内求和：

$$S_{0,0:k-1} = [\sum_{m=0}^{s/k-1} \delta_m^0 A_m^0, \sum_{m=0}^{s/k-1} \delta_m^0 A_m^0 \dots | \sum_{m=0}^{s/k-1} \delta_m^0 A_m^1, \sum_{m=0}^{s/k-1} \delta_m^0 A_m^1 \dots | \dots | \sum_{m=0}^{s/k-1} \delta_m^0 A_m^{k-1}, \sum_{m=0}^{s/k-1} \delta_m^0 A_m^{k-1} \dots \sum_{m=0}^{s/k-1} \delta_m^0 A_m^{k-1}]$$

同理得

$$S_{0,k:2k-1}, S_{0,2k:3k-1} \dots S_{0,t'-k-t:t'-1}$$

$$\text{除以batchSize (s/k) 得到梯度: } \nabla W_{0,0:k-1} = \frac{1}{s/k} \cdot S_{0,0:k-1}$$

同理得到其余的梯度

隐藏层梯度

为方便隐藏层和输出层的梯度都用 δ 表示

对于隐藏层第l个神经元，第m个样本的梯度为： $\delta_m^l = \phi'(Z_m^l) \cdot \sum_{j=0}^{t''-1} \delta_m^{output,j} \cdot W_{j,l}$

其中：

- $\phi'(Z_m^l)$ 是激活函数在 Z_m^l 处的导数
- Z_m^l 是隐藏层第l个神经元的加权输入（激活前）
- W 是隐藏层到输出层的权重

在密文下

$$\frac{\partial L_m}{\partial Z_m^l} = \sum_{j=0}^{t''-1} \frac{\partial L_m}{\partial O_m^j} \frac{\partial O_m^j}{\partial A_m^l} \frac{\partial A_m^l}{\partial Z_m^l}$$

即

$$\sum_{j=0}^{t''-1} \delta_m^j W_{j,l} \phi'(Z_m^l)$$

在密文下计算：

$$\begin{aligned}
R_{0,0:k-1} &= [\delta_0^0 W_{0,0}, \delta_1^0 W_{0,0} \dots \delta_{s/k-1}^0 W_{0,0} | \delta_0^0 W_{0,1}, \delta_1^0 W_{0,1} \dots \delta_{s/k-1}^0 W_{0,1} | \dots | \delta_0^0 W_{0,k-1}, \delta_1^0 W_{0,k-1} \dots \delta_{s/k-1}^0 W_{0,k-1}] \\
R_{1,0:k-1} &= [\delta_0^0 W_{1,0}, \delta_1^0 W_{1,0} \dots \delta_{s/k-1}^0 W_{1,0} | \delta_0^0 W_{1,1}, \delta_1^0 W_{1,1} \dots \delta_{s/k-1}^0 W_{1,1} | \delta_0^0 W_{1,k-1}, \delta_1^0 W_{1,k-1} \dots \delta_{s/k-1}^0 W_{1,k-1}] \\
R_{2,0:k-1} &= [\delta_0^0 W_{2,0}, \delta_1^0 W_{2,0} \dots \delta_{s/k-1}^0 W_{2,0} | \delta_0^0 W_{2,1}, \delta_1^0 W_{2,1} \dots \delta_{s/k-1}^0 W_{2,1} | \delta_0^0 W_{2,k-1}, \delta_1^0 W_{2,k-1} \dots \delta_{s/k-1}^0 W_{2,k-1}] \\
R_{3,0:k-1} &= [\delta_0^0 W_{3,0}, \delta_1^0 W_{3,0} \dots \delta_{s/k-1}^0 W_{3,0} | \delta_0^0 W_{3,1}, \delta_1^0 W_{3,1} \dots \delta_{s/k-1}^0 W_{3,1} | \delta_0^0 W_{3,k-1}, \delta_1^0 W_{3,k-1} \dots \delta_{s/k-1}^0 W_{3,k-1}] \\
&\vdots \\
R_{t''-1,0:k-1} &= [\delta_0^0 W_{t''-1,0}, \delta_1^0 W_{t''-1,0} \dots \delta_{s/k-1}^0 W_{t''-1,0} | \delta_0^0 W_{t''-1,1}, \delta_1^0 W_{t''-1,1} \dots \delta_{s/k-1}^0 W_{t''-1,1} | \delta_0^0 W_{t''-1,k-1}, \delta_1^0 W_{t''-1,k-1} \dots \delta_{s/k-1}^0 W_{t''-1,k-1}]
\end{aligned}$$

上述t''个密文 $\oplus$ 得：

$$[\sum_{j=0}^{t''-1} \delta_0^j W_{j,0}, \sum_{j=0}^{t''-1} \delta_1^j W_{j,0} \dots \sum_{j=0}^{t''-1} \delta_{s/k-1}^j W_{j,0} | \sum_{j=0}^{t''-1} \delta_0^j W_{j,1}, \dots \sum_{j=0}^{t''-1} \delta_{s/k-1}^j W_{j,1} | \dots | \sum_{j=0}^{t''-1} \delta_0^j W_{j,k-1}, \sum_{j=0}^{t''-1} \delta_1^j W_{j,k-1} \dots \sum_{j=0}^{t''-1} \delta_{s/k-1}^j W_{j,k-1}]$$

激活函数的导数

$$\phi'(Z_0^0), \phi'(Z_1^0) \dots \phi'(Z_{s/k-1}^0) | \phi'(Z_0^1), \phi'(Z_1^1) \dots \phi'(Z_{s/k-1}^1) | \dots | \phi'(Z_0^{k-1}), \phi'(Z_1^{k-1}) \dots \phi'(Z_{s/k-1}^{k-1})$$

对应相乘可以得到隐藏层前k个神经元的梯度 $\Gamma_{0:k-1}^h$

同理我们可以得到隐藏层所有神经元的梯度 $\Gamma_{k:2k-1}^h, \Gamma_{2k:3k-1}^h$

共有t'/k个

隐藏层权重梯度计算

对于隐藏层权重 $W_{i,f}$ （连接输入第f个特征到隐藏层第i个神经元），其梯度为： $\frac{\partial \bar{L}}{\partial W_{i,f}} = \frac{1}{s/k} \sum_{m=0}^{s/k-1} \delta_{hidden,m}^i \cdot X_m[f]$

隐藏层的梯度计算结果如下

$$\Gamma_{0:k-1}^h = [\delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \delta_0^1, \delta_1^1 \dots \delta_{s/k-1}^1 | \dots | \delta_0^{k-1}, \delta_1^{k-1} \dots \delta_{s/k-1}^{k-1}]$$

利用掩码提取出上式中k个隐藏层神经元的梯度如下：

$$\begin{aligned}
\Gamma_0^h &= \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \dots \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 \\
\Gamma_1^h &= \delta_0^1, \delta_1^1 \dots \delta_{s/k-1}^1 | \delta_0^1, \delta_1^1 \dots \delta_{s/k-1}^1 | \dots \delta_0^1, \delta_1^1 \dots \delta_{s/k-1}^1 \\
&\vdots \\
\Gamma_{k-1}^h &= \delta_0^{k-1}, \delta_1^{k-1} \dots \delta_{s/k-1}^{k-1} | \delta_0^{k-1}, \delta_1^{k-1} \dots \delta_{s/k-1}^{k-1} | \dots \delta_0^{k-1}, \delta_1^{k-1} \dots \delta_{s/k-1}^{k-1}
\end{aligned}$$

每一个分别与对应的特征相乘

$$\begin{aligned}
& \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \dots \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 \\
& \quad \odot \\
& [X_0[0], \dots, X_{s/k-1}[0] | X_0[1], \dots, X_{s/k-1}[1] | \dots | X_0[k-1], \dots, X_{s/k-1}[k-1]] \\
& \quad \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \dots \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 \\
& \quad \odot \\
& [X_0[k], \dots, X_{s/k-1}[k] | X_0[k+1], \dots, X_{s/k-1}[k+1] | \dots | X_0[2k-1], \dots, X_{s/k-1}[2k-1]] \\
& \quad \vdots \\
& \quad \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 | \dots \delta_0^0, \delta_1^0 \dots \delta_{s/k-1}^0 \\
& \quad \odot \\
& [X_0[t-k-1], \dots, X_{s/k-1}[t-k-1] | X_0[t-k], \dots, X_{s/k-1}[t-k] | \dots | X_0[t-1], \dots, X_{s/k-1}[t-1]]
\end{aligned}$$

每一项组内求和再求平均，得

$$\begin{aligned}
& [\nabla W_{0,0}, \nabla W_{0,0}, \nabla W_{0,0} \dots \nabla W_{0,0} | \nabla W_{0,1} \dots \nabla W_{0,1} | \nabla W_{0,k-1} \dots \nabla W_{0,k-1}] \\
& [\nabla W_{0,k}, \nabla W_{0,k}, \nabla W_{0,k} \dots \nabla W_{0,k} | \nabla W_{0,k+1} \dots \nabla W_{0,k+1} | \nabla W_{0,2k-1} \dots \nabla W_{0,2k-1}] \\
& \quad \vdots \\
& [\nabla W_{0,t-k-1}, \nabla W_{0,t-k-1}, \nabla W_{0,t-k-1} \dots \nabla W_{0,t-k-1} | \nabla W_{0,t-k} \dots \nabla W_{0,t-k} | \nabla W_{0,t-1} \dots \nabla W_{0,t-1}]
\end{aligned}$$

对每一个神经元重复上述操作即可求得所有梯度